

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
S1A_IW_SLC__1SDV_20250616T130237_20250616T130306_059672_0768D1_F57D
S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 899/1

Bridge Chainage – 932+120

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – VUP

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

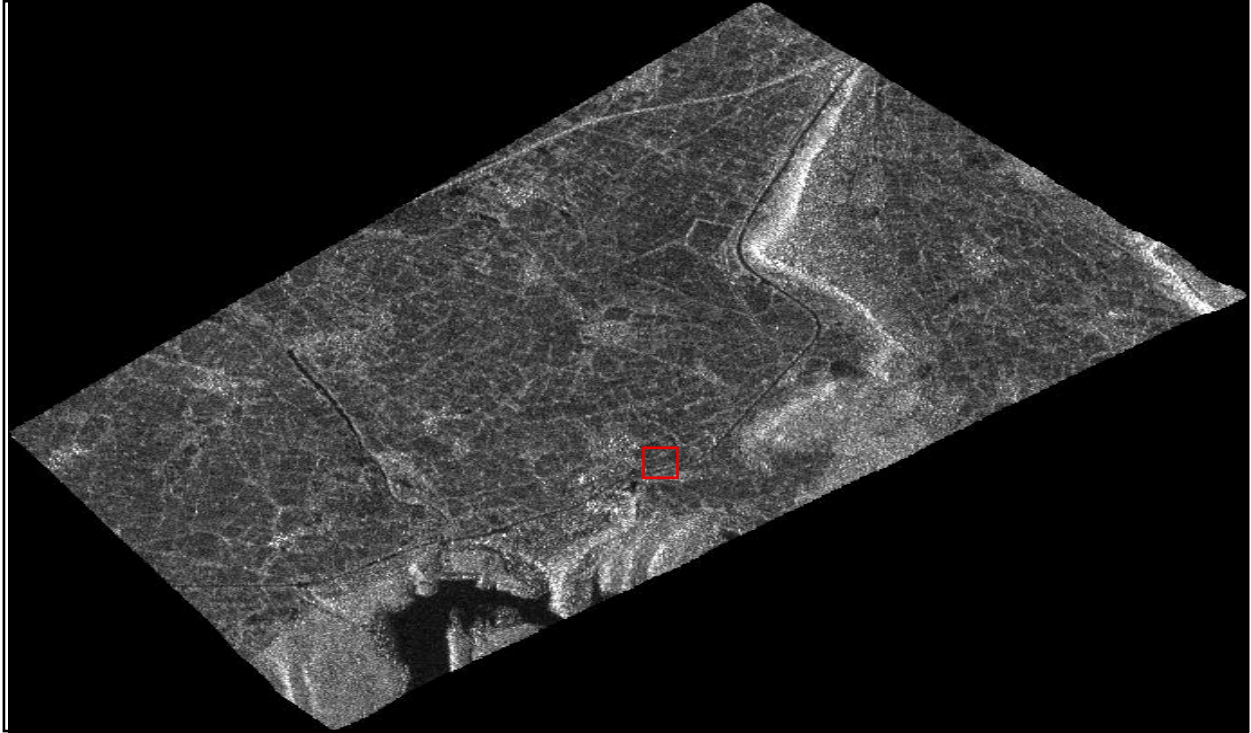
Flight Direction - DESCENDING

Polarization – VV+VH

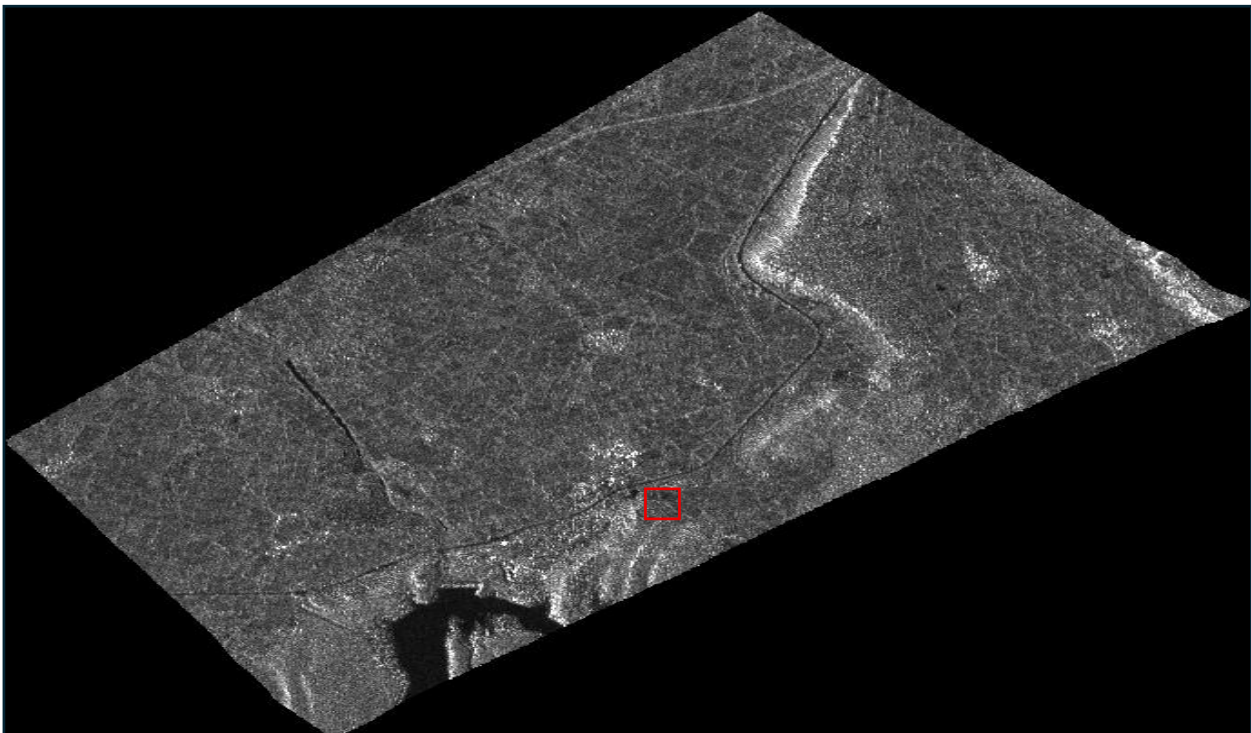
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

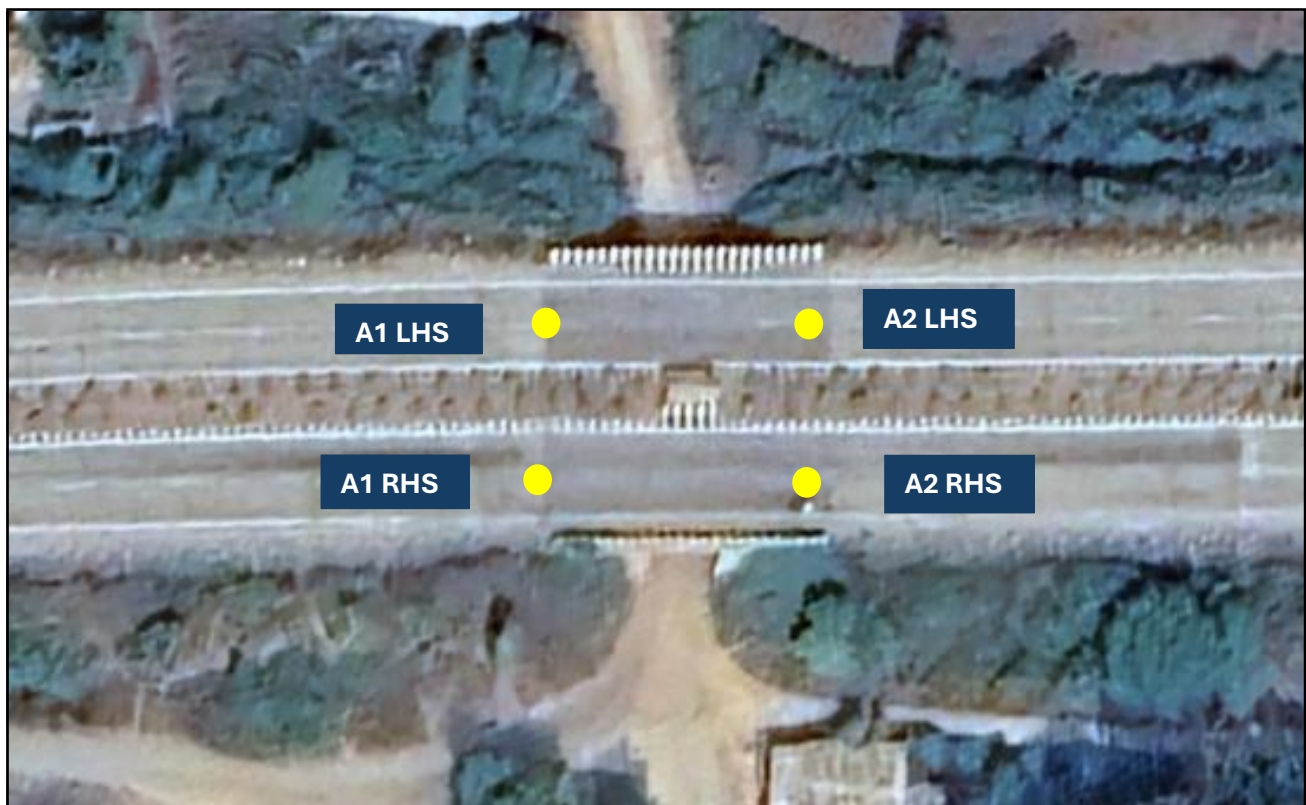


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

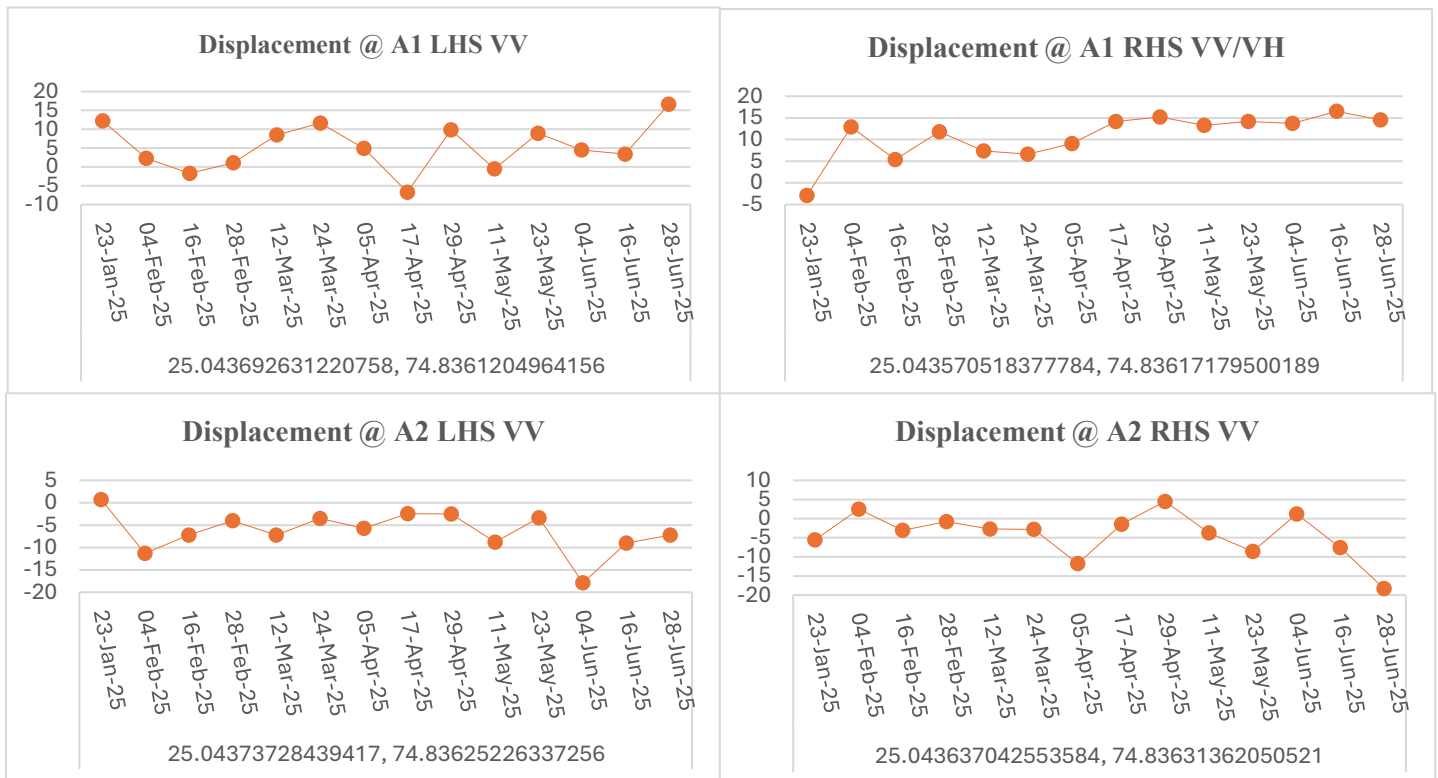
VUP – CH-932+120

Length – 22.99M

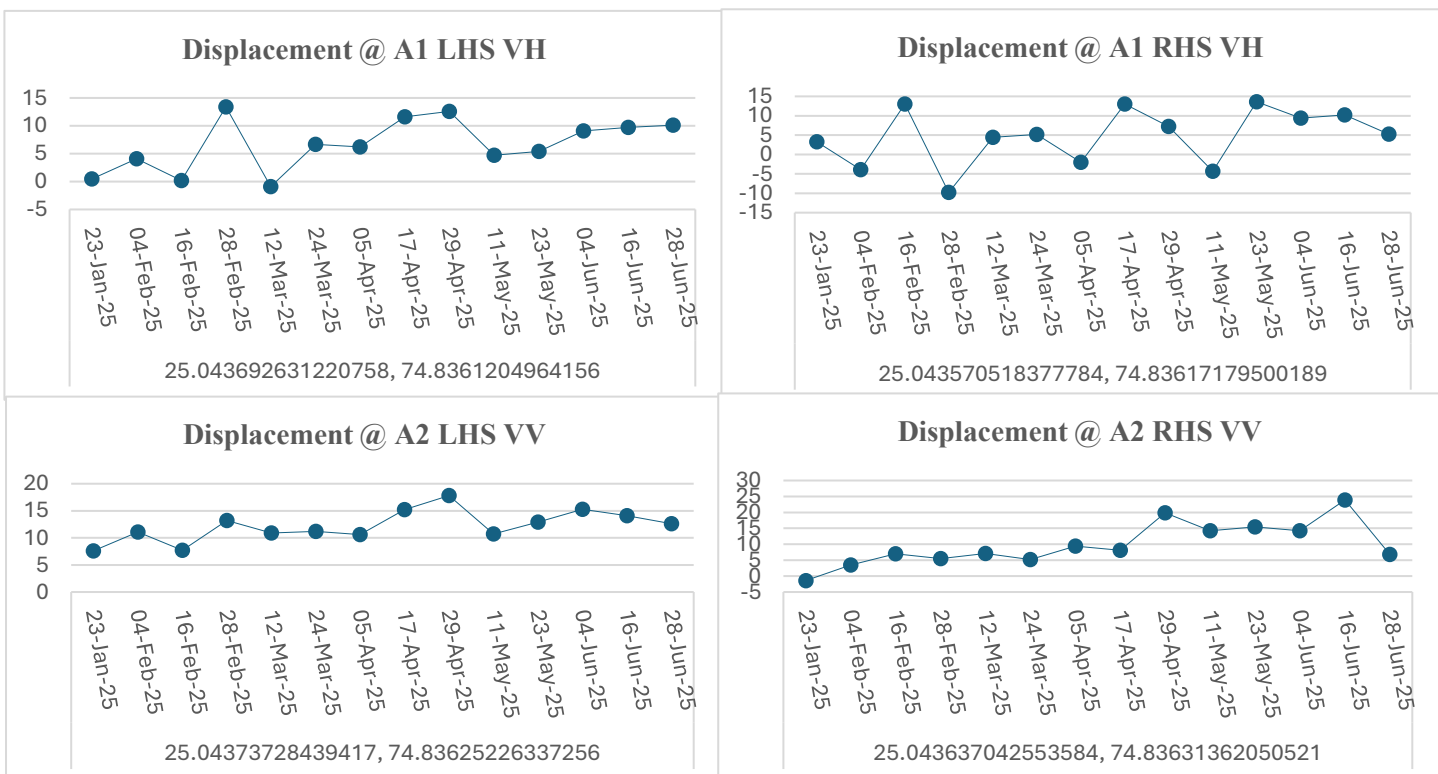


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1	Abutment 1		
A2	Abutment 2		
LHS	Left Hand Side		
RHS	Right Hand Side		

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



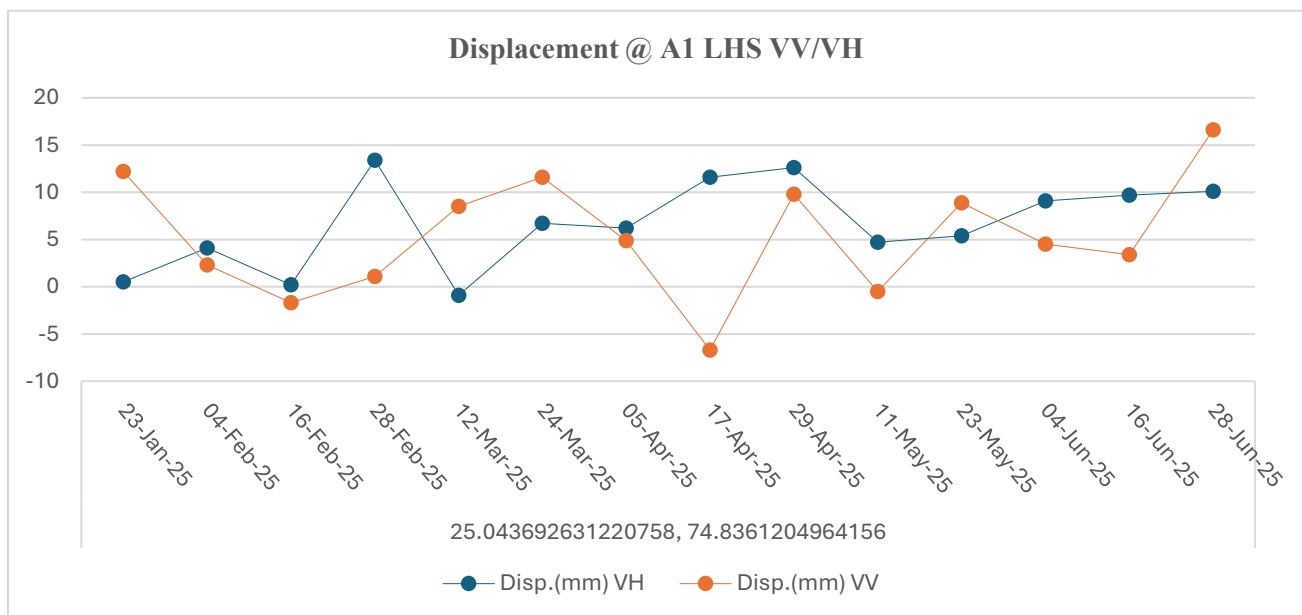
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.043692631220758, 74.8361204964156)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement at A1 LHS shows moderate variability with alternating uplift and settlement phases.
- **Horizontal Displacement (VH)** Horizontal displacement displays lower variability compared to VV, remaining predominantly positive throughout the monitoring period.

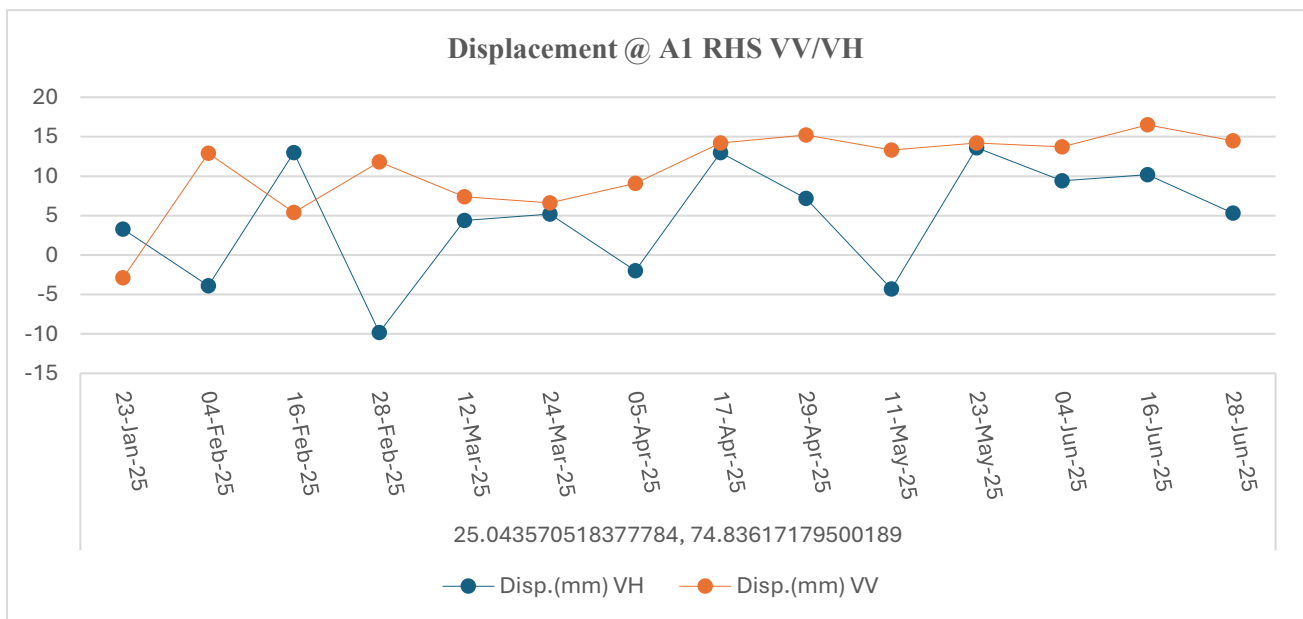
Critical Events & Anomalies:

- **February 2025:** VH rises to +4 mm, while VV decreases to +2 mm, showing a redistribution of displacement from vertical to horizontal components.
- **April 2025:** VV rebounds strongly to +9 mm, and VH remains stable (+12 mm), reflecting rapid vertical recovery.
- **May-June 2025:** Both VH and VV remain consistently positive, with VV increasing to +16 mm and VH stabilising around +8–10 mm, indicating improving structural response and reduced instability.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.043570518377784, 74.83617179500189)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement shows moderate variability with frequent oscillations, ranging roughly between –10 mm and +14 mm.
- **Vertical displacement (VV)** Vertical displacement at A1 RHS exhibits a predominantly positive (uplift-dominated) behaviour throughout the monitoring period.

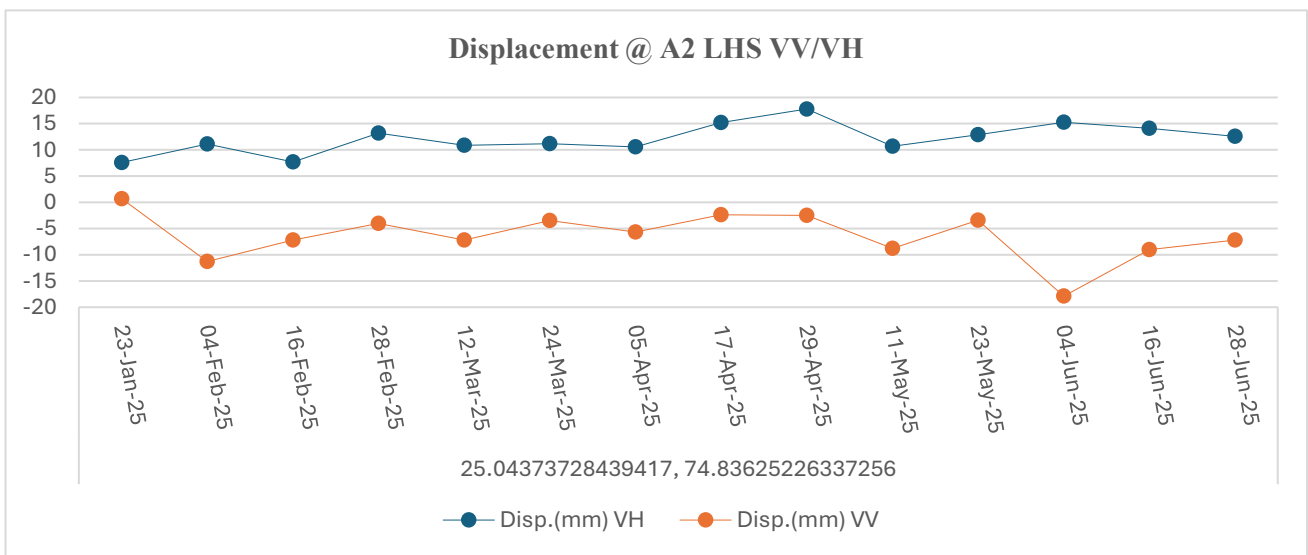
Critical Events & Anomalies:

- **January 2025:** VH begins at +3 mm, while VV is slightly negative (–3 mm), indicating minor initial settlement with limited lateral movement.
- **March 2025:** VH recovers to +4–5 mm, while VV stabilises around +7 mm, suggesting rapid lateral recovery.
- **May-June 2025:** VH recovers and stabilises between +9 mm and +10 mm, while VV reaches its maximum uplift before slightly reducing to +14 mm, indicating overall structural stabilisation.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.04373728439417, 74.83625226337256)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement shows a consistently positive and relatively stable pattern, ranging roughly between +7 mm and +18 mm.
- **Vertical Displacement (VV)** Vertical displacement at A2 LHS is predominantly negative throughout the monitoring period, indicating persistent settlement.

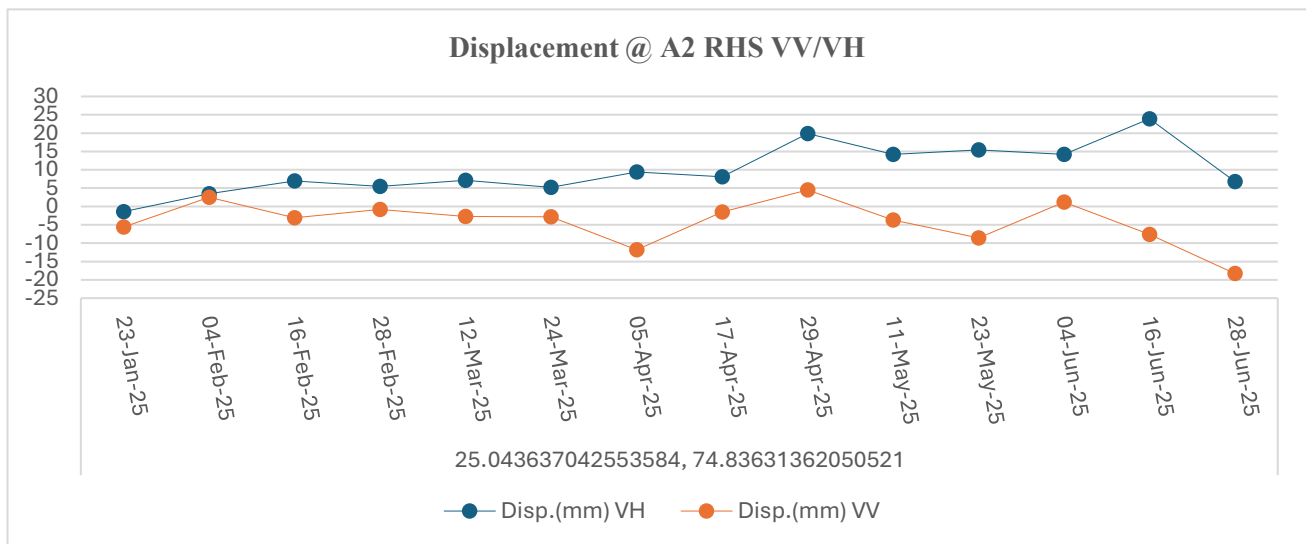
Critical Events & Anomalies:

- **January 2025:** VH begins at +7 mm, while VV is slightly negative (–2 mm), indicating initial settlement with moderate lateral displacement.
- **June 2025:** VV partially recovers to –7 mm, while VH remains stable (+12 mm), confirming persistent settlement with stable horizontal displacement.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.043637042553584, 74.83631362050521)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** Vertical displacement at A2 RHS is predominantly negative, indicating persistent settlement throughout most of the monitoring period.
- **Horizontal (VH) displacement** Horizontal displacement shows a strongly positive and progressively increasing trend, ranging roughly between +2 mm and +24 mm.

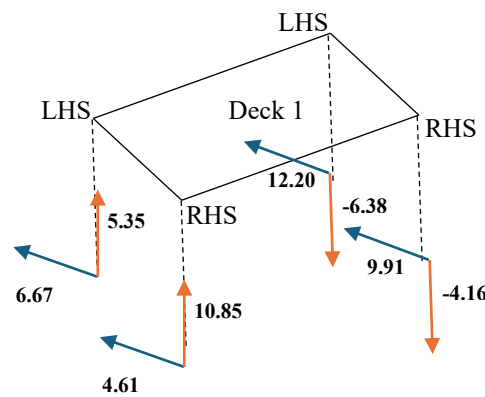
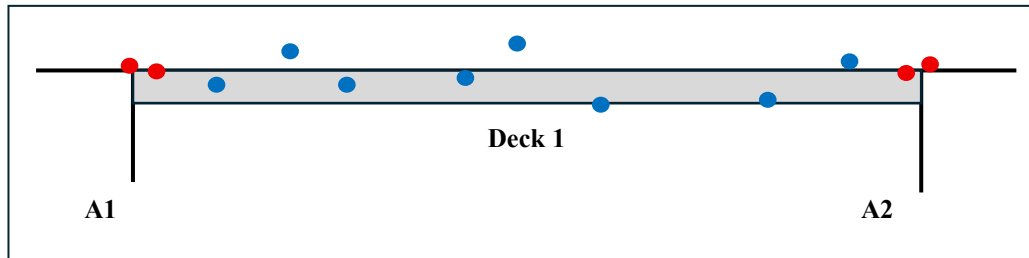
Critical Events & Anomalies:

- **January 2025:** VH starts near +2 mm, while VV is negative (−7 mm), indicating initial settlement accompanied by limited lateral displacement.
- **May 2025:** VH remains elevated, while VV declines again to −6 to −9 mm, suggesting renewed settlement under sustained lateral movement
- **June 2025:** VV reaches its maximum settlement, while VH drops to +6 mm, highlighting a late-stage vertical degradation with partial lateral relaxation.

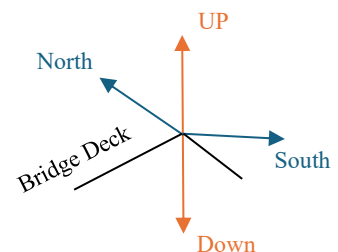
ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Bridge Deck Slab Analysis



Vertical Displacement Behaviour:

- The bridge deck slab exhibits clearly non-uniform vertical displacement, indicating differential vertical response between the LHS and RHS regions. The maximum vertical uplift is observed at the central RHS support, reaching approximately +10.85 mm, representing the dominant upward response of the deck.

Horizontal Displacement Behaviour:

- Horizontal displacement shows strong directional variability across the deck width, reflecting torsional deformation behaviour. The maximum horizontal displacement occurs at the central deck region, reaching approximately +12.20 mm, indicating significant lateral movement of the deck core.

CONCLUSION:

- The bridge deck slab demonstrates a torsion-dominated deformation response, characterized by significant lateral displacement combined with asymmetric vertical uplift and localized settlement. The concentration of maximum horizontal movement near the deck core, together with uplift at both supports, highlights complex deck-support interaction. Overall, the observed displacement magnitudes and asymmetry indicate structurally significant behaviour, warranting continued monitoring and correlation with loading, bearing response, and boundary constraints.